POLITECNICO DI TORINO

Technology of Metallic Materials A.Y. 2025-26

Mechanical Engineering Course

Casto irons

Giovanni Maizza
DISAT

Lecture 13

1

POLITECNICO DI TORINO

Cast Iron

- **General Properties**: 2-4.5% Carbon and 1-3% Si.
- Ferrite or pearlite matrix together with graphite as second phase (slow cooling)
- Low melting point, ease of casting, very fluid, low shrinkage, easily machinable.
- Low impact resistance and ductility.
- **Types of Cast Iron**:
  - **White cast iron**
  - **Gray cast iron**
  - **Malleable cast iron**
  - **Ductile cast iron**

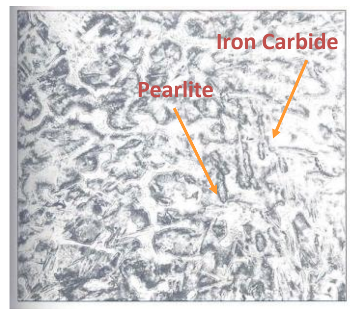
9-48

2

POLITECNICO DI TORINO

White Cast iron

- Carbon is present in the form of Iron Carbide instead of graphite up on solidification.
- Fractured surface appears white and crystalline.
- Low C (2.5 – 3%) and Si (0.5 – 1.5%) content.
- Excellent wear resistance
- High hardness
- High brittleness



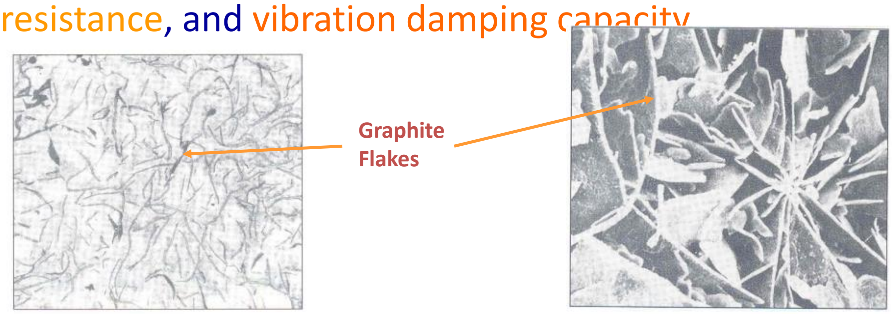
9-49 Courtesy of central Foundry

3

POLITECNICO DI TORINO

Gray Cast Iron

- C= 2.5 – 4% and Si=1 – 3% (graphite former).
- C exceeds the amount that can be dissolved in austenite and precipitate as graphite flakes.
- Fractured surface appears gray.
- Excellent machinability, hardness and wear resistance, and vibration damping capacity



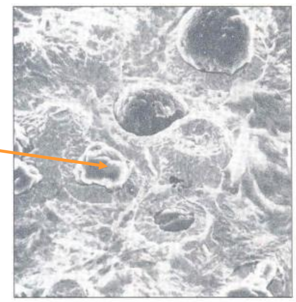
9-50

4

POLITECNICO DI TORINO

Ductile Cast iron

- Has processing advantages of cast iron and engineering advantages of steel.
- Good fluidity, castability, machinability, and wear resistance.
- High strength, toughness, ductility and hardenability (due to spherical nodules of graphite).
- 3-4% C and 1.8 – 2.8 % Si and low impurities.
- Bull's eye type microstructure.



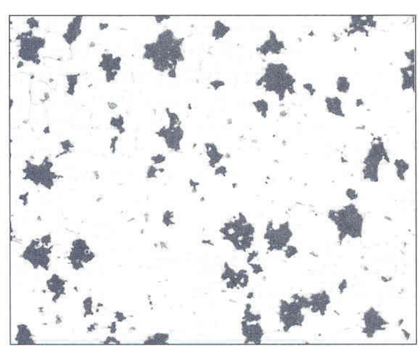
9-51

5

POLITECNICO DI TORINO

Malleable Cast Iron

- 2-2.6 % C and 1.1 – 1.6% Si.
- White cast iron is heated in malleablizing furnace to dislocate carbide into graphite.
- Irregular nodules of graphite are formed.
- Good castability, machinability, moderate strength, toughness and uniformity.



6

POLITECNICO
DI TORINO

end